

Outline

1.Executive Summary

2.Introduction

3.Methodology

4.Results

5.Conclusion

6.Appendix

Executive Summary

- Project Background:

- Predicting Falcon 9 first stage landing success
- SpaceX cost advantage due to reusable first stage
- Cost implications for competing companies

- Key Questions:

- Factors influencing successful rocket landing
- Impact of rocket variables on landing success rate
- Optimal conditions for SpaceX to achieve best results

Introduction

- Data collection from SpaceX API and Wikipedia
- Data wrangling for machine learning preparation
- Exploratory Data Analysis (EDA) using visualization and SQL
- Interactive visual analytics with Folium and Plotly Dash
- Predictive analysis using classification models

Methodology Overview

- 1.Data Collection
- 2.Data Wrangling
- 3.Exploratory Data Analysis (EDA)
- 4.Interactive Map Creation
- 5.Dashboard Development
- 6.Predictive Analysis

Data Collection

- SpaceX REST API
 - Endpoints: api.spacexdata.com/v4/
 - Data normalized into CSV format
- Web Scraping (Wikipedia)
 - BeautifulSoup used for HTML parsing
 - Data extracted and normalized into CSV

Data Wrangling

- Conversion of landing outcomes to binary labels:
 - 1: Successful landing
 - 0: Unsuccessful landing
- Process:
 - Calculate launches per site
 - Determine mission outcomes per orbit type
 - Create landing outcome labels
 - Calculate success rates
 - Export processed dataset as CSV

Exploratory Data Analysis (EDA)

- Visualization Techniques:
 - Scatter plots: Flight Number vs. Payload Mass, Launch Site, etc.
 - Bar graphs: Mean vs. Orbit
 - Line graphs: Success Rate vs. Year
- SQL Queries:
 - Unique launch sites
 - Payload mass statistics
 - Landing outcomes analysis
 - Booster version performance

Predictive Analysis (Classification)

- Process:

- Data preparation and splitting
- Model selection and training
- Hyperparameter tuning with GridSearchCV
- Model evaluation
- Feature engineering and algorithm tuning

- Best Performing Model: Decision Tree Classifier

- Accuracy on test data: 83.33%

Interactive Map with Folium

- Visualization of launch sites on an interactive map
- Features:
 - Circle markers for launch sites
 - Color-coded markers for success/failure
 - Distance calculations to landmarks using Haversine formula
- Findings:
 - Launch sites proximity to coastlines
 - Distance from cities and infrastructure

Dashboard with Plotly Dash

- Interactive dashboard built with Flask and PlotlyDash
- Hosted on PythonAnywhere for 24/7 access
- Key Components:
 - Scatter graph: Outcome vs. Payload Mass for different Booster Versions
 - Pie chart: Total launches by site

Predictive Analysis (Classification)

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Conclusion

- Decision Tree Classifier: Best algorithm for this dataset
- Payload weight significantly affects landing success
- Launch success rates improve over time
- KSC LC-39A: Most successful launch site
- Orbits GEO, HEO, SSO, ES-L1 have highest success rates

Key Results

1.EDA Insights:

- 1.Higher flight numbers correlate with increased success rates
- 2.Certain orbits (GEO, HEO, SSO, ES-L1) show better success rates
- 3.Payload mass impacts success differently for various orbits

2.Launch Site Performance:

- 1.KSC LC-39A achieved highest success rate (76.9%)

3.Payload Impact:

- 1.Lower weight payloads generally have higher success rates

4.Yearly Trend:

- 1.Success rate has been increasing since 2013

Appendix

- Haversine formula for distance calculations
- Custom modules:
 - ADGGoogleMaps (for coordinate retrieval and mapping)
 - ADGSQLSERVER (for database interactions)
- PythonAnywhere for 24/7 dashboard hosting